

AQEEL ABBAS KHAN

Software Engineer | Full-Stack & AI

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PROFESSIONAL SUMMARY

Software Engineer with production experience delivering AI-powered web applications and full-stack systems. Builds React frontends, Node.js and FastAPI backends, and data layers on PostgreSQL, MongoDB, and Supabase (BaaS). Hands-on with REST APIs, RAG pipelines, LLM agents, NLP, Docker, Git, and Agile delivery. Open to remote, onsite, hybrid, and freelance Software Engineer roles.

Core Competencies: Software Engineering, Full-Stack Development, React.js, Node.js, Express.js, FastAPI, Flask, Python, JavaScript, REST APIs, PostgreSQL, MongoDB, MySQL, Supabase, RAG, LangChain, LangGraph, LLMs, NLP, PyTorch, Docker, Azure, Git

WORK EXPERIENCE

Software Developer (AI & Full Stack) | GenTech | Pakistan

December 2025 – Present

- Developed and maintained production web and AI systems using Python, FastAPI, React.js, PostgreSQL, and Supabase Backend-as-a-Service for authentication, database, and API workflows.
- Contributed to SoulPlus AI, a Call Center Management System, and BiedBuddy, integrating REST APIs and AI-powered modules in a team-based Agile environment.
- Designed Retrieval-Augmented Generation (RAG) pipelines and conversational AI agents with LangChain, LangGraph, Large Language Models (LLMs), and Supabase PostgreSQL with pgvector.
- Built backend services for data ingestion, validation, transformation, and storage to support reliable performance on growing datasets.
- Implemented secure REST APIs for communication between internal services and external systems.
- Used Git for version control and Docker for consistent development and deployment environments.

PROJECTS

AdGenie | AI Advertisement Generation Platform | Final Year Project

React.js, Node.js, FastAPI, MongoDB, PyTorch, LLMs

- Built an end-to-end AI platform connecting a React.js frontend, Node.js REST APIs, and a FastAPI PyTorch inference service for prompt-driven advertisement generation.
- Designed generative workflows for marketing creatives and editable AI-generated content.
- Presented the project at Namal Ideathon and Innovative Pakistan 2026, a national-level technology showcase.

SoulPlus AI | Production AI Assistant

LangGraph, LangChain, RAG, FastAPI, Supabase (BaaS), PostgreSQL, pgvector

- Developed a production AI assistant with LangGraph multi-step reasoning, tool orchestration, and memory-aware conversations.
- Implemented RAG over a Supabase vector database for contextual knowledge retrieval.
- Built a reusable document pipeline that extracted, cleaned, chunked, and embedded 100+ PDF and DOCX files into the agent knowledge base.
- Delivered a FastAPI backend for AI inference and orchestration.

Automated Essay Scoring | NLP Research Application

BERT, GPT, PyTorch, Hugging Face Transformers, Python, NLP

- Fine-tuned BERT for multi-dimensional essay scoring across grammar, coherence, vocabulary, and content.
- Used GPT to generate human-like feedback from model outputs.
- Designed the NLP pipeline for preprocessing, tokenization, and evaluation against scoring benchmarks.

TixTrader | Full-Stack Ticket Resale Marketplace

React.js, Node.js, Express.js, MongoDB, JWT

- Built a full-stack marketplace with buyer, seller, and admin workflows.
- Implemented JWT authentication, role-based access control, ticket listings, and purchase APIs.
- Designed backend architecture for event management and ticket transactions.

NearBuy | Location-Based E-Commerce Marketplace

MongoDB, Express.js, React.js, Node.js, Tailwind CSS

- Developed a MERN marketplace with geolocation-based vendor discovery, filtering, and nearby shop results.
- Implemented a responsive storefront connecting local buyers and vendors across device sizes.

Kohinoor Fertilizers | Client E-Commerce Platform

React.js, Supabase (BaaS), Admin Dashboard

- Delivered a production e-commerce website with product listing, catalog management, and an admin dashboard.
- Used Supabase as a Backend-as-a-Service for authentication, database, and backend APIs, with reusable React components.

Obesity Prediction System | Machine Learning

Python, Pandas, NumPy, Scikit-learn

- Trained and evaluated supervised machine learning models to predict obesity levels from lifestyle and health data.
- Performed data cleaning, feature engineering, model comparison, and statistical evaluation, with a dashboard for data insights.

TECHNICAL SKILLS

Languages: Python, JavaScript, C++

Frontend: React.js, HTML5, CSS3, Tailwind CSS, Responsive Design, REST API Integration

Backend: Node.js, Express.js, FastAPI, Flask, REST APIs, Supabase (Backend-as-a-Service)

Databases: PostgreSQL, MongoDB, MySQL, Supabase Postgres, Vector Databases (pgvector)

AI / Machine Learning: Large Language Models (LLMs), RAG, AI Agents, LangChain, LangGraph, PyTorch, TensorFlow, Scikit-learn, NLP, Hugging Face, BERT, GPT

Tools and Cloud: Git, GitHub, Docker, Microsoft Azure, Jira, Postman, Linux

EDUCATION

Bachelor of Science in Computer Science | Namal University, Mianwali, Pakistan

August 2022 – May 2026

Relevant Coursework: Data Structures and Algorithms, Software Engineering, Database Systems, Web Development, Cloud Computing, Operating Systems, Artificial Intelligence, Machine Learning, Natural Language Processing, Deep Learning

CERTIFICATIONS AND ACHIEVEMENTS

- **Innovative Pakistan 2026** — National-level technology showcase participant; presented AdGenie at The University of Faisalabad.
- **Namal Ideathon and Innovation Expo** — Presented Final Year Project AdGenie at the university innovation event.
- **Data Science with Python** — LetsUpgrade. Additional: Machine Learning Fundamentals; Python Programming.